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MAHATMA GANGHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR / IMPROVEMENT / REAPPEARANCE EXAMINATIONS, APRIL2026

SECOND SEMESTER

Multi-Disciplinary Course (MDC) –MG2MDCBTS100-BIOLOGICAL FOUNDATIONS FOR HEALTH AND WELLNESS

(2024 ADMISSION ONWARDS)

Part A

Answer all questions

Each carries one mark

1.What are alleles? [K]/[CO3]

Ans: Different versions of a gene

2.Which part of the flower develops into the fruit after fertilization? [K]/[CO3]

Ans: Ovary

3.Which type of muscle is found in heart? [K]/[CO2]

Ans: Cardiac muscle

4.What is the function of spinal cord? [K]/[CO5]

Ans: Transmits messages between the brai and body.

5.Where does fertilization typically occur? [K]/[CO2]

Ans: Fallopian tube

Part B

Short Answer Questions

Answer any 5 questions

Each question carries 2 marks

6.What dietary changes are recommended in managing Type 2 diabetes? [U]/[CO4]

Managing Type 2 diabetes involves consuming balanced, portion-controlled meals, reducing added sugars, refined carbohydrates and saturated fats.

7.Define mental health disorder. [K]/[CO5]

A mental health disorder is a clinically significant disturbance in an individual's cognition, emotional regulation, or behaviour.

8.How does genetics influence an individual's health? [K]/[CO1]

Genetics influence health by dictating how the body functions, metabolizes nutrients, and responds to environmental factors, directly affecting risks for diseases.

9.Give four examples for calcium rich vegetables. [K]/[CO1]

Dark leafy greens such as Kale, Broccoli, Collard greens, Bok Choy are good sources of vegetables.

10.What is circulatory system? [K]/[CO2]

The circulatory system is an extensive network of organs and vessels that pumps and carries blood throughout the body, supplying cells with oxygen and nutrients while removing waste materials. It consists of the heart, blood vessels such as arteries, veins, capillaries and blood, acting as the body's primary transport system for energy, growth, and repair.

11.What is the highest level of biological organization? [U]/[CO1]

The largest level of biological organization is the biosphere**,** which encompasses all living organisms on Earth, interacting with the abiotic components of the environment.

12.What are the two types of immunity in the human body? [K]/[CO5]

The two main types of immunity in the human body are innate immunity and adaptive immunity**.** Innate immunity provides an immediate, non-specific response to pathogens, while adaptive immunity offers a tailored, long-lasting defence that develops over time.

Part C

Short Essay Questions

Answer any 5 questions

Each question carries 4 marks

13.Describe the nutrient molecules required during lactation? [U]/[CO1]

The nutrient molecules required during lactation are
1.Protein**:** Essential for the synthesis of milk proteins and for the growth of the baby. The High-quality sources include lean meats, dairy, eggs, beans and lentils.
2.Fatty acids: Docosahexaenoic acid (DHA) is a long-chain polyunsaturated fatty acid crucial for the infant's brain and eye development. It is found in fatty fish, walnuts, and flaxseeds.
3.Calcium**:** Supports bone strength for both mother and baby. Found in dairy, leafy greens and tofu.
4.Iodine**:** Crucial for the baby's brain and nervous system development. Sources include the seafood.
5.Choline**:** Needed to support infant brain development.
6.Vitamin A: Important for immune function and vision development.
7.Vitamin D: Helps in calcium absorption and immunity. Supplementation is often recommended if the mother has low sun exposure.
8.Vitamin B12**:** Essential for neurological development.

14.Explain healthy eating pyramid. [K]/[CO4]

The components of a Healthy Eating Pyramid are
1.Base Layer (Eat Most): Whole grains (oats, brown rice, wholemeal bread) provide necessary fibre and energy.
2.Second Layer (Eat More): A wide variety of vegetables, salad, and fruits provide vitamins and minerals.
3.Third Layer (Eat Moderately): Dairy products for calcium, and protein alternatives (fish, eggs, beans, nuts, poultry) for muscle health.
4.Top Layer (Eat Least): Fats, oils, salt, and sugar should be limited to prevent obesity and health issues.

15.What is central dogma of molecular biology? [K]/[CO3]

The components of the Central Dogma
The process consists of three main stages
A. Replication: DNA acts as a template to produce more DNA, ensuring genetic information is passed on.
B. Transcription
The process where a specific segment of DNA (gene) is copied into a single-stranded RNA molecule (mRNA).

C. Translation: The mRNA molecule is read by ribosomes in the cytoplasm, translating the RNA sequence into a specific sequence of amino acids to form a protein.

16.Describe the importance of the thyroid gland in metabolism. [K]/[CO5]

Important metabolic functions of thyroid gland are
1.Energy Regulation: Thyroid hormones regulate the overall metabolic rate.
2.Nutrient Metabolism: These hormones stimulate the metabolism of fats, carbohydrates, and proteins, affecting both their synthesis and breakdown.
3.Thermogenesis: The thyroid helps control body temperature by influencing energy expenditure, often raising temperature through increased cellular activity.
4.Organ Activity: They regulate key bodily functions, including heart rate, cholesterol levels, and digestive tract movement.

17.Name the major organs of the digestive system and their functions. [K]/[CO2]

The parts of the digestive system are

- Oral Cavity**:** Begins digestion through chewing (mechanical) and saliva (chemical).
- Pharynx: Directs food and liquids to the esophagus while protecting the airway.
- Esophagus: Transports food from the pharynx to the stomach via muscular waves called peristalsis.
- Stomach: Temporarily stores food, breaks it down with gastric acids and enzymes, and converts it into a thin, watery mixture.
- Small Intestine: The main site for digestion and nutrient absorption, consisting of the duodenum, jejunum, and ileum.
- Large Intestine: Absorbs water, electrolytes, and vitamins from remaining food residue, forming feces.

The digestive Organs are

- Salivary Glands: Produce saliva to lubricate food and begin carbohydrate digestion.
- Liver: Produces bile to break down fats and detoxifies harmful substances.
- Gallbladder: Stores and concentrates bile until needed in the small intestine.
- Pancreas: Produces enzymes (pancreatic juice) to digest fats, proteins, and carbohydrates, along with blood-sugar regulating hormones.

18.What is dialysis and when it is needed? [K]/[CO2]

Dialysis is a life-saving medical treatment that filters waste, toxins, and excess fluids from the blood when kidneys can no longer function properly. It is typically required in cases of acute kidney injury (temporary) or when chronic kidney disease reaches end-stage renal disease.

19.What are the strategies for maintaining respiratory health? [K]/[CO2]

The strategies for respiratory health
1.Minimize Exposure to Pollutants:
Outdoor: Check air quality reports, avoid exercising near high-traffic areas, and wear a mask on high-pollution days.
Indoor: Reduce allergens by dusting and vacuuming, avoid air fresheners/aerosol sprays, use air purifiers, and keep homes mold-free.
2.Engage in Regular Exercise: Aerobic activity, such as brisk walking, running strengthens the heart and lungs, enhancing their ability to transport oxygen
Deep Breathing Exercises: Helps clear lungs and improves capacity.
3.Maintain a Healthy Lifestyle:
a. Balanced Diet: Consume antioxidant-rich foods to fight oxidative stress.
b. Good Posture: Standing or sitting tall allows for maximum lung expansion.